📘 AZ-104 – LAB 05

Implement Intersite Connectivity

🎯 Lab Objective

The objective of this lab is to understand **how Azure virtual networks communicate**, how connectivity is **enabled and tested**, and how **traffic flow can be controlled** using custom routes.

This lab simulates a **real enterprise scenario** where different business units are isolated but still require controlled communication.

🏗 Architecture Overview

- **CoreServicesVnet** → Core IT services
- **ManufacturingVnet** → Manufacturing workloads
- **VNet Peering** → Private connectivity
- **Network Watcher & PowerShell** → Connectivity testing
- **User Defined Routes (UDR)** → Traffic control via future NVA

🔹 Task 1: Create Core Services VM and VNet

What was done

- Created CoreServicesVM
- Created CoreServicesVnet (10.0.0.0/16)
- Created Core subnet (10.0.0.0/24)

Why this is important

- Provides an endpoint to test connectivity
- Demonstrates VM-first network creation
- Uses private IP-only access for security

🔹 Task 2: Create Manufacturing VM and VNet

What was done

- Created ManufacturingVM
- Created ManufacturingVnet (172.16.0.0/16)
- Created Manufacturing subnet (172.16.0.0/24)

Key concept

- Azure VNets are **isolated by default**
- No communication is allowed unless explicitly configured

🔹 Task 3: Test Connectivity using Network Watcher

Tool used

- **Network Watcher → Connection Troubleshoot**

Result

- Connectivity test failed (Unreachable)

Why this happens

- VNets are isolated
- No routing exists between them

This confirms Azure's **secure-by-default** behavior.

🔹 Task 4: Configure Virtual Network Peering

What was configured

- Peering between CoreServicesVnet and ManufacturingVnet
- Enabled:
  - Virtual network access
  - Forwarded traffic

What peering provides

- Private IP communication
- Low latency

- Microsoft backbone routing
- No VPN gateway required

🔹 Task 5: Test Connectivity using PowerShell

Command used

Test-NetConnection <Private-IP> -Port 3389

Result

- TcpTestSucceeded : True

What this proves

- VNet peering works
- Private routing is functional
- Connectivity is verified at OS level

🔹 Task 6: Create Custom Route (User Defined Route)

Why routing is required

- Enterprises do not allow unrestricted traffic
- All traffic must pass through inspection devices

What was created

- Perimeter subnet (10.0.1.0/24)
- Route table rt-CoreServices
- Custom route directing traffic to **future NVA (10.0.1.7)**

Important concept

"Future NVA" represents a planned firewall or security appliance.

The route ensures traffic **will be forced through security controls** when the NVA is deployed.

🔑 Key Takeaways

- VNets are isolated by default
- VNet peering enables private connectivity
- Network Watcher helps diagnose connectivity issues
- PowerShell provides script-based verification
- User-defined routes override Azure system routes
- Traffic control is critical in enterprise networks

🧠 3️⃣ INTERVIEW QUESTIONS & ANSWERS (VERY IMPORTANT)

Q1: Why can't Azure VNets communicate by default?

A: Azure isolates VNets by default to ensure security and prevent unintended access.

Q2: What is Virtual Network Peering?

A: VNet peering connects two VNets privately, allowing resources to communicate using private IPs over the Microsoft backbone.

Q3: Is VPN Gateway required for VNet peering?

A: No. VNet peering does not require a VPN gateway or public IPs.

Q4: What is Network Watcher used for?

A: It is used to monitor, diagnose, and troubleshoot network connectivity issues in Azure.

Q5: What is a User Defined Route (UDR)?

A: A UDR is a custom route that overrides Azure system routes to control traffic flow.

Q6: What is a Network Virtual Appliance (NVA)?

A: An NVA is a VM acting as a firewall, proxy, or security device to inspect and control traffic.

Q7: Why use a "future NVA" in the lab?

A: To demonstrate routing architecture without deploying expensive firewall appliances.

Q8: At what level are route tables applied?

A: Route tables are applied at the subnet level.